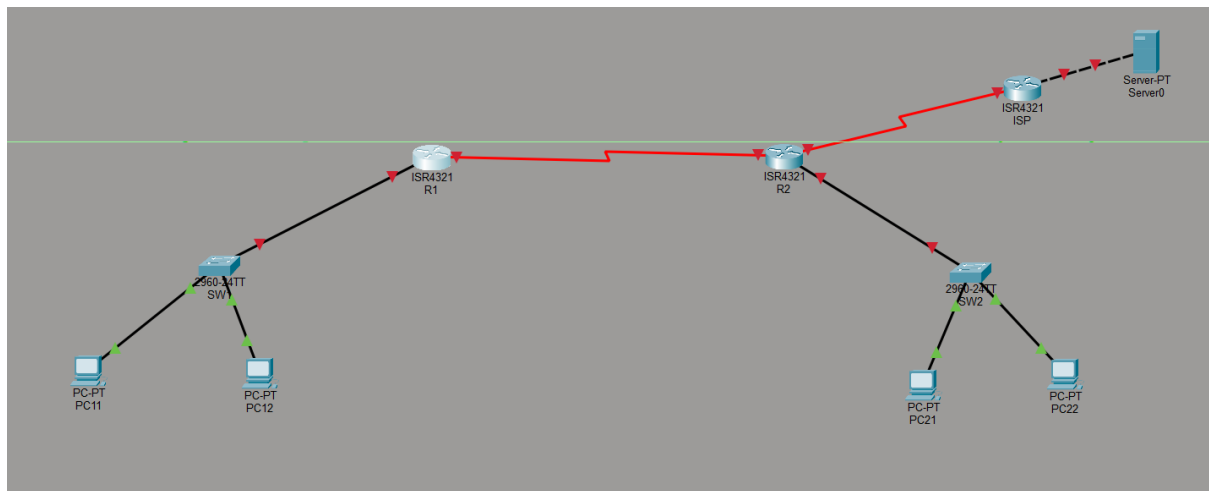




```

Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#username abdou secret isg
R1(config)#line cons
R1(config)#line console 0
R1(config-line)#login local
R1(config-line)#exit
R1(config)#line vty
R1(config)#line vty
R1(config)#line vty
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#exit
R1(config)#

```

R1#show ipv6 interface

link local:

24|FFFE|24 7aja 9rawha fil cour

To make it local it has to be among: FE8 or FE9 or FEA or FEB

```
R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int g0/0/0
R1(config-if)#ipv6 add FE80::1 link-local
R1(config-if)#exit
R1(config)#int s0/1/0
R1(config-if)#ipv6 add FE90::1 link-local
R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#wr
Building configuration...
[OK]
R1#ipv6 int brief
      ^
% Invalid input detected at '^' marker.

R1#show ipv6 int brief
GigabitEthernet0/0/0      [up/up]
    FE80::1
    2001:DB8:AC:1::1
GigabitEthernet0/0/1      [administratively down/down]
    unassigned
Serial0/1/0               [down/down]
    FE90::1
    2001:DB8:AC:3::1
Serial0/1/1               [administratively down/down]
    unassigned
Serial0/2/0               [administratively down/down]
    unassigned
Serial0/2/1               [administratively down/down]
    unassigned
Vlan1                     [administratively down/down]
    unassigned
R1#
```

```

R1(config)#line vty
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#exit
R1(config)#
R1(config)#
R1(config)#enable secret class
R1(config)#no ip domain-lookup
R1(config)#int g0/0/0
R1(config-if)#description CONNECT TO SW1
R1(config-if)#ipv6 address 2001:db8:ac:1::1/64
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

R1(config-if)#exit
R1(config)#int s0/1/0
R1(config-if)#desc
R1(config-if)#description CONNECT TO R2
R1(config-if)#ipv6 add 2001:db8:ac:3::1/64
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
R1(config-if)#
R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#wr
Building configuration...
[OK]
R1#show ipv6 interface

```

```

R1#show ipv6 int brief
GigabitEthernet0/0/0      [up/up]
    FE80::1
    2001:DB8:AC:1::1
GigabitEthernet0/0/1      [administratively down/down]
    unassigned
Serial0/1/0               [down/down]
    FE90::1
    2001:DB8:AC:3::1
Serial0/1/1               [administratively down/down]
    unassigned
Serial0/2/0               [administratively down/down]
    unassigned
Serial0/2/1               [administratively down/down]
    unassigned
Vlan1                     [administratively down/down]
    unassigned
R1#
R1#

```

Do the same for R2 :

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#username abdou secret isg
R2(config)#line console 0
R2(config-line)#login local
R2(config-line)#exit
R2(config)#line vty 0 4
R2(config-line)#login local
R2(config-line)#exit
R2(config)#enable secret class
R2(config)#do wr
Building configuration...
[OK]
R2(config)#int g0/0/0
R2(config-if)#des
R2(config-if)#description CONNECT TO SW2
R2(config-if)#ipv6 add 2001:db8:ac:2::1/64
R2(config-if)#ipv6 add FEA0::1 link-local
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

R2(config-if)#
```

```
R2(config-if)#exit
R2(config)#int s0/1/0
R2(config-if)#des
R2(config-if)#description CONNECT TO R1
R2(config-if)#ipv6 add 2001:db8:ac:3::2/64
R2(config-if)#ipv6 add FE90::2 link-local
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

R2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#wr
Building configuration...
[OK]
R2#
```

```
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int s0/1/1
R2(config-if)#desc
R2(config-if)#description CONNECT TO ISP
R2(config-if)#ipv6 add 2001:db8:ac:4::1/64
R2(config-if)#ipv6 add FEB0::1 link-local
R2(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/1, changed state to down
R2(config-if)#do
```

```

ND reachable time is 30000 milliseconds
R2#show ipv6 int brief
GigabitEthernet0/0/0      [up/up]
    FE80::1
    2001:DB8:AC:2::1
GigabitEthernet0/0/1      [administratively down/down]
    unassigned
Serial0/1/0               [up/up]
    FE90::2
    2001:DB8:AC:3::2
Serial0/1/1               [down/down]
    FE80::1
    2001:DB8:AC:4::1
Serial0/2/0               [administratively down/down]
    unassigned
Serial0/2/1               [administratively down/down]
    unassigned
Vlan1                     [administratively down/down]
    unassigned
R2#

```

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ISP
ISP(config)#username abdou secret ISG
ISP(config)#line con 0
ISP(config-line)#login local
ISP(config-line)#exit
ISP(config)#line vty 0 4
ISP(config-line)#login local
ISP(config-line)#exit
ISP(config)#enable secret class
ISP(config)#no ip domain-lookup
ISP(config)#ip domain name ISG.com.tn
ISP(config)#crypto key generate RSA
The name for the keys will be: ISP.ISG.com.tn
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

ISP(config)#

```

```

ISP(config)#line vty 0 4
*Mar 1 1:3:50.62: %SSH-5-ENABLED: SSH 1.99 has been enabled
ISP(config-line)#transport input all

```

```
ISP#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ISP(config)#ip ssh version 2
ISP(config)#ip ssh auth
ISP(config)#ip ssh authentication-retries 2
ISP(config)#ip ssh tim
ISP(config)#ip ssh time-out 60
ISP(config)#end
ISP#
%SYS-5-CONFIG_I: Configured from console by console

ISP#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 60 secs; Authentication retries: 2
ISP#
```

```
ISP#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ISP(config)#int s0/1/1
ISP(config-if)#des
ISP(config-if)#description CONNECT TO R2
ISP(config-if)#ipv6 add 2001:db8:ac:4::2/64
ISP(config-if)#ipv6 add FE80::2 link-local
ISP(config-if)#no shutdown
```

```
ISP(config-if)#exit
ISP(config)#int g0/0/0
ISP(config-if)#dec
ISP(config-if)#desc
ISP(config-if)#description CONNECT TO SERVER
ISP(config-if)#ipv6 add 2001:db8:ac:5::1/64
ISP(config-if)#ipv6 add FE80::2 link-local
ISP(config-if)#no shutdown

ISP(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

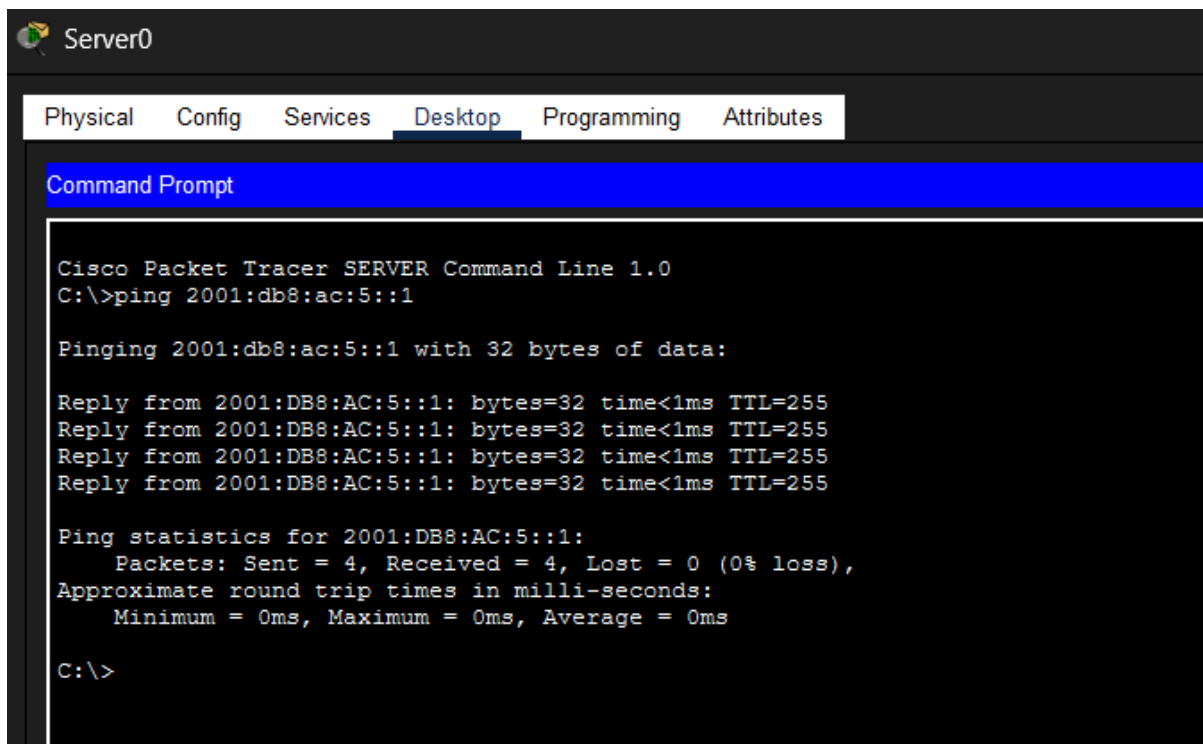
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

ISP(config-if)#end
ISP#
%SYS-5-CONFIG_I: Configured from console by console

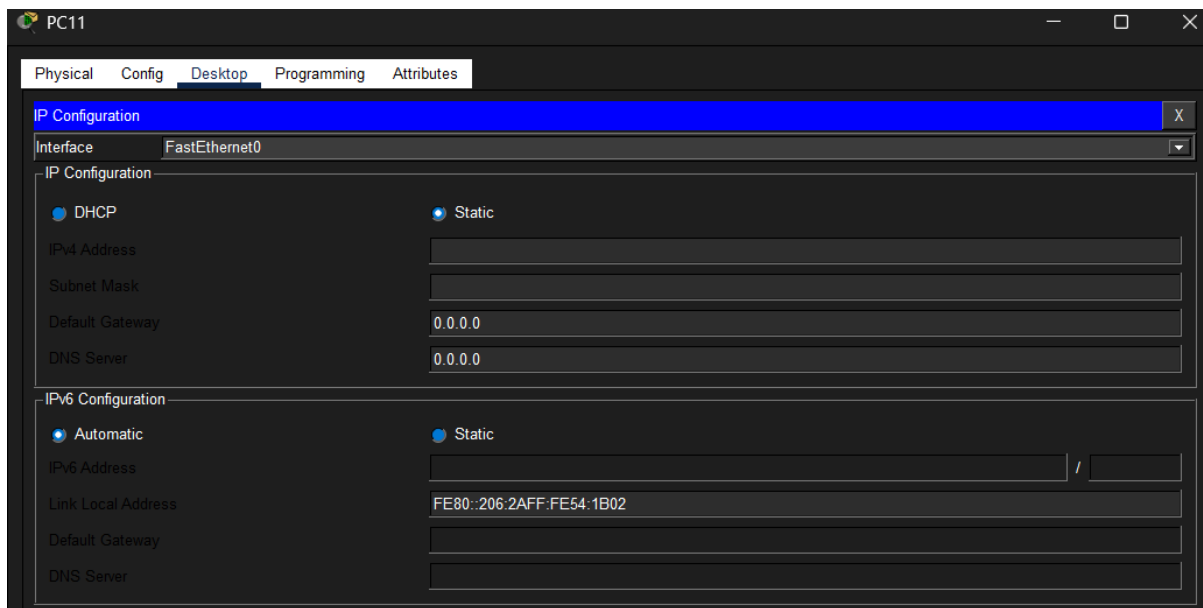
ISP#wr
Building configuration...
[OK]
ISP#show ipv6 int
GigabitEthernet0/0/0 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::2
  No Virtual link-local address(es):
  Global unicast address(es):
    2001:DB8:AC:5::1, subnet is 2001:DB8:AC:5::/64
  Joined group address(es):
    FF02::1
    FF02::1:FF00:1
    FF02::1:FF00:2
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachable are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds
Serial0/1/1 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::2
  No Virtual link-local address(es):
  Global unicast address(es):
    2001:DB8:AC:4::2, subnet is 2001:DB8:AC:4::/64
  Joined group address(es):
    FF02::1
--More--
```



Fun fact dns server is of google



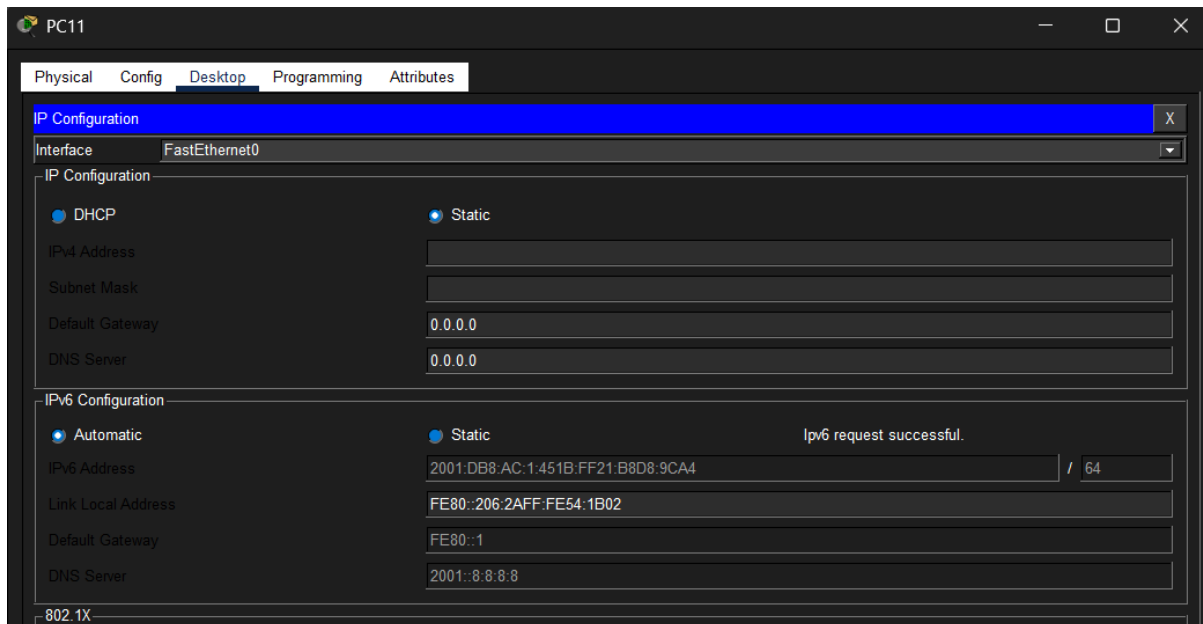
Put pc 11 automatic



User Access Verification

Username: abdou
Password:

```
R1>en
Password:
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ipv6 dhcp
% Incomplete command.
R1(config)#ipv6 dhcp po
R1(config)#ipv6 dhcp pool LAN1
R1(config-dhcpv6)#add prefix 2001:db8:ac:1::/64
R1(config-dhcpv6)#dns-se
R1(config-dhcpv6)#dns-server 2001::8:8:8:8
R1(config-dhcpv6)#domain-name ISG.com.tn
R1(config-dhcpv6)#exit
R1(config)#int g0/0/0
R1(config-if)#ipv6 dhcp server LAN1
R1(config-if)#ipv6 nd mana
R1(config-if)#ipv6 nd managed-config-flag
R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
R1#
```

```
R2>en
Password:
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ipv6 dhcp
% Incomplete command.
R2(config)#ipv6 dhcp pool LAN2
R2(config-dhcpv6)#add prefix 2001:db8:ac:2::/64
R2(config-dhcpv6)#dns se
R2(config-dhcpv6)#dns-s
R2(config-dhcpv6)#dns-server 2001::8:8:8:8
R2(config-dhcpv6)#domain-name ISG.com.tn
R2(config-dhcpv6)#exit
R2(config)#int g0/0/0
R2(config-if)#ipv6 dhcp server LAN2
R2(config-if)#ipv6 nd managed-config-flag
R2(config-if)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#wr
Building configuration...
[OK]
R2#
```

Pc21

